Chapter 3: Crosstalk

Outline

- Mutual Inductance and Mutual Capacitance
- Inductance and Capacitance Matrix
- Field Simulators
- Crosstalk-Induced Noise
- Simulating Crosstalk Using Equivalent Circuit Models
- Crosstalk-Induced Flight Time and Signal Integrity Variations
- Crosstalk Trends
- Termination of Odd- and Even-Mode Transmission Line Pairs
- Minimization of Crosstalk
- Additional Examples

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1. Introduction

1.1 Influences of Crosstalk

(chap3.pdf, p.3)

Disadvantages: 1. Crosstalk changes the **impedance** and **propagation velocity** of a line → adversely affects system-level timing and signal integrity 2. Crosstalk **induces noise** onto neighboring lines → degrades signal integrity and reduces noise margins

Influence Factors: Data patterns, line-to-line spacing, and switching rates.

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2. Mutual Inductance and Mutual Capacitance

2.1 Mutual Inductance

(chap3.pdf, p.4)

- The coupling of current via the **magnetic field** is modeled by mutual inductance L_m
- L_m injects a **voltage noise** onto the victim line proportional to the **rate of change of current** on the driver line:

$$V_{\text{noise},L_m} = L_m \frac{dI_{\text{driver}}}{dt}$$

(chap3.pdf, p.4)

2.2 Mutual Capacitance

(chap3.pdf, p.5)

- The coupling due to the **electric field** is modeled by mutual capacitance C_m
- C_m injects a **current noise** onto the victim line proportional to the **rate of change of voltage** on the driver line:

$$I_{\text{noise},C_m} = C_m \frac{dV_{\text{driver}}}{dt}$$

(chap3.pdf, p.5)

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3. Inductance and Capacitance Matrix

3.1 Introduction

(chap3.pdf, p.6)

The inductance and capacitance matrices are collectively called the **transmission line matrices**.

3.2 Inductance Matrix

(chap3.pdf, p.6)

For an N-conductor system:

$$\text{Inductance matrix} = \begin{bmatrix} L_{11} & L_{12} & \cdots & L_{1N} \\ L_{21} & L_{22} & & \\ \vdots & & \ddots & \\ L_{N1} & & & L_{NN} \end{bmatrix}$$

(chap3.pdf, p.6)

- L_{NN}: **self-inductance** of line N
- L_{MN}: **mutual inductance** between lines M and N

3.3 Capacitance Matrix

(chap3.pdf, p.7)

$$\text{Capacitance matrix} = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1N} \\ C_{21} & C_{22} & & \\ \vdots & & \ddots & \\ C_{N1} & & & C_{NN} \end{bmatrix}$$

(chap3.pdf, p.7)

- C_{NN}: **total capacitance seen by line** N = capacitance to ground + all mutual capacitances to other lines
- C_{NM}: **mutual capacitance** between conductors N and M

3.4 Example 3.1: Two-Conductor TX Line Matrices

(chap3.pdf, p.8)

For a two-conductor system, the charge-voltage relation is:

$$\begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

(chap3.pdf, p.8)

Key relationships: - $C_{11} = C_{1g} + C_m$ (self-capacitance: capacitance to ground + mutual) - $C_{12} = C_{21} = -C_m$ (mutual capacitance is negative in this convention)

C_{11} is obtained by setting $V_2 = 0$ (grounding line 2).

The inductance matrix is:

$$\text{Inductance matrix} = \begin{bmatrix} L_{11} & L_{12} \\ L_{21} & L_{22} \end{bmatrix}$$

(chap3.pdf, p.8)

> **Figure description (p.8):** Cross-section diagram showing two adjacent signal traces above a ground plane. Mutual capacitor C_m connects the two traces; C_{1g} and C_{2g} are individual capacitances to the ground plane.

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4. Field Simulators

4.1 Electrostatic (2D) Simulators

(chap3.pdf, p.9)

- **Advantages:** Very easy to use; fast computation
- **Disadvantages:**
  - Only simulate relatively simple geometries
  - Based on static electric field calculations
  - Do not calculate frequency-dependent effects (internal inductance, skin-effect resistance)
  - Alternative methods exist for frequency-dependent R and L

4.2 Full-Wave (3D) Simulators

(chap3.pdf, p.10)

- **Advantages:**
  - Simulate complex 3D geometries
  - Predict frequency-dependent losses, internal inductance, dispersion, radiation, and most other EM phenomena
- **Disadvantages:**
  - Very difficult to use; simulations take **hours or days**
  - Output is often S-parameters, which are **not directly useful** for digital interconnect simulations

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5. Crosstalk-Induced Noise

5.1 NEXT and FEXT Categories

(chap3.pdf, p.11)

Near-End Crosstalk (NEXT) — Backward Crosstalk:

$$I_{\text{near}} = I(L_m) + I_{\text{near}}(C_m)$$

(chap3.pdf, p.11)

The near-end current is the **sum** of inductance-induced and capacitance-induced currents.

Far-End Crosstalk (FEXT) — Forward Crosstalk:

$$I_{\text{far}} = I_{\text{far}}(C_m) - I(L_m)$$

(chap3.pdf, p.11)

The far-end current is the **difference** of capacitance-induced and inductance-induced currents.

> **Figure description (p.11):** Two parallel traces (Line 1 = driver, Line 2 = victim). Driver current flows right on Line 1. Mutual capacitance C_m (or C_{12}) connects the two lines. Arrows on Line 2 show: $I(L_m)$ directed backward (near end), $I_{\text{near}}(C_m)$ directed backward, and $I_{\text{far}}(C_m)$ directed forward (far end).

5.2 Timing of Crosstalk Pulses

(chap3.pdf, p.12)

- **Near-end crosstalk:** begins at $t = 0$, duration = 2 TD (twice the electrical length of the line)
- **Far-end crosstalk:** occurs at $t = \text{TD}$, duration $\approx$ signal rise or fall time

The magnitude and shape depend on the amount of coupling and the termination.

> **Figure description (p.12):** Three snapshots of the coupled two-line system showing wave propagation. (Top) Signal injected at beginning: near-end crosstalk pulse propagates backward, far-end crosstalk pulse propagates forward. (Middle) Signal at midpoint (t = TD/2): partial crosstalk pulses visible. (Bottom) Signal at end (t = TD): far-end crosstalk arrives at far end.

5.3 Matched Terminations — When $T_r < 2\,\text{TD}$

(chap3.pdf, p.13)

For matched terminations ($R = Z_0$ at both ends):

$$\text{TD} = X\sqrt{LC}$$

(chap3.pdf, p.13)

where X = line length, L and C = self-inductance and capacitance per unit length.

Case 1 (driver terminated, victim terminated at both ends — $R = Z_0$ everywhere):

Near-end magnitude:

$$A = \frac{V(\text{input})}{4}\left(\frac{L_M}{L} + \frac{C_M}{C}\right)$$

(chap3.pdf, p.13)

Far-end magnitude:

$$B = -\frac{V(\text{input})\,X\sqrt{LC}}{2T_r}\left(\frac{L_M}{L} - \frac{C_M}{C}\right)$$

(chap3.pdf, p.13)

Case 2 (victim only terminated at far end — $R = Z_0$ at far end only):

$$A = \frac{V(\text{input})}{4}\left(\frac{L_M}{L} + \frac{C_M}{C}\right), \quad B = \frac{1}{2}C, \quad C = -\frac{V(\text{input})\,X\sqrt{LC}}{T_r}\left(\frac{L_M}{L} - \frac{C_M}{C}\right)$$

(chap3.pdf, p.13)

Case 3 (victim not terminated at either end):

$$A = \frac{V(\text{input})}{2}\left(\frac{L_M}{L} + \frac{C_M}{C}\right), \quad B = \frac{V(\text{input})\,X\sqrt{LC}}{2T_r}\left(\frac{L_M}{L} - \frac{C_M}{C}\right), \quad C = \frac{V}{4}\left(\frac{L_M}{L} - \frac{C_M}{C}\right)$$

(chap3.pdf, p.13)

> **Figure description (p.13):** Left column: three circuit diagrams (Case 1, 2, 3) with different termination configurations. Right column: corresponding near-end V(near) and far-end V(far) waveforms for each case, showing trapezoidal pulses with widths $2X\sqrt{LC}$ (near end) and $X\sqrt{LC}$ (far end) labeled.

5.4 Matched Terminations — When $T_r > 2\,\text{TD}$

(chap3.pdf, p.14)

- **Near-end crosstalk** fails to reach its maximum amplitude → multiply by $2\,\text{TD}/T_r$:

$$V_{\text{near,adjusted}} = V_{\text{near,max}} \times \frac{2\,\text{TD}}{T_r}$$

(chap3.pdf, p.14)

- **Far-end crosstalk** equations do **not** need adjustment.

Key conclusions: - Near-end magnitude is **independent of length** when $T_r < 2\,\text{TD}$ (long-line case) - Far-end magnitude **always depends on rise time and length**

5.5 Mismatched Terminations — Only Victim Mismatched

(chap3.pdf, p.15)

When the victim termination $R \neq Z_0$, the crosstalk is modified by a reflection factor:

$$V_x = V_{\text{crosstalk}} \left(1 + \frac{R - Z_o}{R + Z_o}\right)$$

(chap3.pdf, p.15)

where: - V_x: adjusted crosstalk at near or far end of victim line - R: termination impedance - Z_0: characteristic impedance of TX line - $V_{\text{crosstalk}}$: matched-case crosstalk value

> Note: Near-end noise is modified by the reflection coefficient at the near end; far-end noise is modified by the reflection coefficient at the far end.

5.6 Example 3.2: Matched Terminations

(chap3.pdf, p.17–18)

Given: $Z_0 \approx 70\,\Omega$, termination resistors $= 70\,\Omega$, $V(\text{input}) = 1.0\,\text{V}$, $T_r = 100\,\text{ps}$, $X = 2\,\text{in.}$

$$L/\text{in.} = \begin{bmatrix} 9.869\,\text{nH} & 2.103\,\text{nH} \\ 2.103\,\text{nH} & 9.869\,\text{nH} \end{bmatrix}, \quad C/\text{in.} = \begin{bmatrix} 2.051\,\text{pF} & -0.239\,\text{pF} \\ -0.239\,\text{pF} & 2.051\,\text{pF} \end{bmatrix}$$

(chap3.pdf, p.17)

Near-end crosstalk (Case 1):

$$V(1) = \frac{V(\text{input})}{4} \left[\frac{L_M}{L} + \frac{C_M}{C}\right] = \frac{1}{4}\left(\frac{2.103\,\text{nH}}{9.869\,\text{nH}} + \frac{0.239\,\text{pF}}{2.051\,\text{pF}}\right) = 0.082\,\text{V}$$

(chap3.pdf, p.18)

Far-end crosstalk (Case 1):

$$V(2) = -\frac{V(\text{input})\,X\sqrt{LC}}{2T_r} \left[\frac{L_M}{L} - \frac{C_M}{C}\right] = -0.137\,\text{V}$$

(chap3.pdf, p.18)

5.7 Example 3.3: Mismatched Victim Terminations

(chap3.pdf, p.19)

Given: Same setup as Example 3.2, but $R_1 = 45\,\Omega$ (near end), $R_2 = 100\,\Omega$ (far end)

$$Z_o = \sqrt{\frac{L_{11}}{C_{11}}} = \sqrt{\frac{9.869\,\text{nH}}{2.051\,\text{pF}}} = 69.4\,\Omega$$

(chap3.pdf, p.19)

$$V(1) = 0.082\left(1 + \frac{45 - 69.4}{45 + 69.4}\right) = 0.0645\,\mathrm{V}$$

(chap3.pdf, p.19)

$$V(2) = -0.137\left(1 + \frac{100 - 69.4}{100 + 69.4}\right) = -0.162\,\mathrm{V}$$

(chap3.pdf, p.19)

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6. Simulating Crosstalk Using Equivalent Circuit Models

6.1 N-Segment Coupled Line Model

(chap3.pdf, p.20)

An N-segment SPICE model for coupled lines includes: - Line a (upper): L_a, C_a, R_a, G_a per segment - Line b (lower): L_b, C_b, R_b, G_b per segment - Mutual coupling capacitors C_m between lines at each node - Inductive coupling represented by coupling factor K

> **Figure description (p.20):** Schematic of N-segment coupled-line model. Upper trace has series L_a, shunt $C_a \parallel G_a$, series R_a per segment. Lower trace similarly. Between the two traces: C_m capacitors at each node. Mutual inductor coupling $K_1 \ldots K_N$ shown between inductors of the two lines.

6.2 Inductive Coupling Factor K

(chap3.pdf, p.21)

The mutual inductance is modeled as a coupling factor:

$$K = \frac{L_{12}}{\sqrt{L_{11}L_{22}}}$$

(chap3.pdf, p.21)

where L_{12} = mutual inductance between lines 1 and 2.

6.3 Example 3.4: Creating a Coupled Line Model

(chap3.pdf, p.21–23)

Given: 5 in. long coupled pair, rise time = 100 ps

$$C/\text{in.} = \begin{bmatrix} 2\,\mathrm{pF} & -0.1\,\mathrm{pF} \\ -0.1\,\mathrm{pF} & 2\,\mathrm{pF} \end{bmatrix}, \quad L/\text{in.} = \begin{bmatrix} 9\,\mathrm{nH} & 0.7\,\mathrm{nH} \\ 0.7\,\mathrm{nH} & 9\,\mathrm{nH} \end{bmatrix}$$

(chap3.pdf, p.21)

Characteristic impedance:

$$Z_o = \sqrt{\frac{L_{11}}{C_{11}}} = \sqrt{\frac{9\,\text{nH}}{2\,\text{pF}}} = 67.09\,\Omega$$

(chap3.pdf, p.22)

Total propagation delay:

$$\text{TD} = \sqrt{L_{11}C_{11}} = \sqrt{(9\,\text{nH})(2\,\text{pF})} = 134\,\text{ps/in.} \times 5\,\text{in.} = 670\,\text{ps}$$

(chap3.pdf, p.22)

Inductive coupling factor:

$$K = \frac{L_{12}}{\sqrt{L_{11}L_{22}}} = \frac{0.7\,\text{nH}}{9\,\text{nH}} = 0.078$$

(chap3.pdf, p.22)

Number of segments:

$$N = \text{segments} = 10\frac{X}{vT_r} = 10\frac{(5\,\text{in.})(134\,\text{ps/in.})}{100\,\text{ps}} = 67$$

(chap3.pdf, p.23)

Per-segment values (multiply by 5/67 in./segment): - $L(N) = 0.67\,\text{nH}$ - $C_{1g}(N) = C_{11}(N) - C_m(N) = 0.1425\,\text{pF}$ - $C_m(N) = 0.0075\,\text{pF}$

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7. Crosstalk-Induced Flight Time and Signal Integrity Variations

7.1 Odd-Mode Analysis

(chap3.pdf, p.24–27)

Definition: Odd mode — two lines carry **equal and opposite** signals: $I_2 = -I_1$, $V_2 = -V_1$

Applying KVL to the coupled inductor circuit ($L_{11} = L_{22} = L_0$):

$$V_1 = L_0\frac{dI_1}{dt} + L_m\frac{dI_2}{dt}$$

(chap3.pdf, p.24)

Substituting odd-mode conditions:

$$V_1 = (L_0 - L_m)\frac{dI_1}{dt}$$

(chap3.pdf, p.25)

Odd-mode inductance:

$$L_{\text{odd}} = L_{11} - L_m = L_{11} - L_{12}$$

(chap3.pdf, p.25)

The odd mode inductance is **reduced**.

Applying KCL with $C_{1g} = C_{2g} = C_0$, for odd mode ($V_2 = -V_1$):

$$I_1 = (C_{1g} + 2C_m)\frac{dV_1}{dt}$$

(chap3.pdf, p.26)

Odd-mode capacitance:

$$C_{\text{odd}} = C_{1g} + 2C_m = C_{11} + C_m$$

(chap3.pdf, p.26)

The odd mode capacitance is **increased**.

Odd-mode impedance and time delay:

$$Z_{\text{odd}} = \sqrt{\frac{L_{\text{odd}}}{C_{\text{odd}}}} = \sqrt{\frac{L_{11} - L_{12}}{C_{11} + C_m}}$$

(chap3.pdf, p.27)

$$\text{TD}_{\text{odd}} = \sqrt{L_{\text{odd}} C_{\text{odd}}} = \sqrt{(L_{11} - L_{12})(C_{11} + C_m)}$$

(chap3.pdf, p.27)

The odd-mode characteristic impedance is **reduced**; time delay effect is not obvious.

7.2 Even-Mode Analysis

(chap3.pdf, p.28–29)

Definition: Even mode — two lines carry **identical** signals: $I_1 = I_2$, $V_1 = V_2$

Substituting even-mode conditions:

$$V_1 = (L_0 + L_m)\frac{dI_1}{dt}$$

(chap3.pdf, p.28)

Even-mode inductance:

$$L_{\text{even}} = L_{11} + L_m$$

(chap3.pdf, p.28)

The even mode inductance is **increased**.

For even-mode capacitance ($V_1 = V_2$, so $V_1 - V_2 = 0$, no current through C_m):

$$C_{\text{even}} = C_0 = C_{11} - C_m$$

(chap3.pdf, p.29)

The even mode capacitance is **reduced**.

Even-mode impedance and time delay:

$$Z_{\text{even}} = \sqrt{\frac{L_{\text{even}}}{C_{\text{even}}}} = \sqrt{\frac{L_{11} + L_{12}}{C_{11} - C_m}}$$

(chap3.pdf, p.29)

$$\text{TD}_{\text{even}} = \sqrt{L_{\text{even}} C_{\text{even}}} = \sqrt{(L_{11} + L_{12})(C_{11} - C_m)}$$

(chap3.pdf, p.29)

The even-mode impedance is **increased**; time delay effect is not obvious.

7.3 Field Patterns of Odd and Even Modes

(chap3.pdf, p.30)

> **Figure description (p.30):** Four diagrams showing field patterns. Top row (electric field): Odd mode — E-field lines bow between the two conductors (increased capacitance); Even mode — E-field lines go symmetrically to ground (decreased capacitance). Bottom row (magnetic field): Odd mode — two separate tight loops (decreased inductance); Even mode — one large combined loop (increased inductance).

7.4 Switching Pattern Effects on Performance

(chap3.pdf, p.31)

- **Odd mode:** $Z_S = Z_0 > Z_{\text{odd}} \rightarrow$ **underdriven** behavior
- **Even mode:** $Z_S = Z_0 < Z_{\text{even}} \rightarrow$ **overdriven** (ringing) behavior

> **Figure description (p.31):** Waveform plot showing V(source) and V(load) for three traces (V1, V2, V3). Even mode (all rising together): rises slowly and settles at 2 V (underdriven staircase). Odd mode (center trace out of phase): overshoot and ringing. Annotation: "Why does odd-mode arrive early?" — because reduced inductance leads to faster wave propagation in odd mode.

7.5 Conclusions on Odd/Even Mode Effects

(chap3.pdf, p.32)

1. **Odd-mode impedance** is always **lower** than the single-line Z_0
2. **Even-mode impedance** is always **higher** than the single-line Z_0
3. Time delay effects for odd/even mode are **not obvious** (opposite effects partially cancel)
4. Crosstalk **will** induce velocity variations in a **microstrip**
5. Crosstalk **will not** induce velocity variations in a **stripline**

7.6 Single-Line Equivalent Model (SLEM) — All Signals In Phase

(chap3.pdf, p.33)

For a three-line bus (lines 1, 2, 3) with all signals in phase (even mode with respect to center line 2):

$$Z_{2,\text{eff}} = \sqrt{\frac{L_{22} + L_{12} + L_{23}}{C_{22} - C_{12} - C_{23}}}$$

(chap3.pdf, p.33)

$$\text{TD}_{2,\text{eff}} = \sqrt{(L_{22} + L_{12} + L_{23})(C_{22} - C_{12} - C_{23})}$$

(chap3.pdf, p.33)

7.7 SLEM — One Signal Out of Phase

(chap3.pdf, p.34)

When line 2 is out of phase with its neighbors (differential mode — center bit different):

Equivalent capacitance of conductor 2: - Even-mode C with 1: $C_{22} - C_{12}$ - Odd-mode C with 3: $C_{22} + C_{23}$ - Equivalent: $C_{22} - C_{12} + C_{23}$

Equivalent inductance of conductor 2: - Even-mode L with 1: $L_{22} + L_{12}$ - Odd-mode L with 3: $L_{22} - L_{23}$ - Equivalent: $L_{22} + L_{12} - L_{23}$

$$Z_{2,\text{eff}} = \sqrt{\frac{L_{22} + L_{12} - L_{23}}{C_{22} - C_{12} + C_{23}}}$$

(chap3.pdf, p.34)

$$\text{TD}_{2,\text{eff}} = \sqrt{(L_{22} + L_{12} - L_{23})(C_{22} - C_{12} + C_{23})}$$

(chap3.pdf, p.34)

7.8 SLEM Conclusions

(chap3.pdf, p.35)

- **Nearest neighbors** have the greatest crosstalk effect; effects fall off **exponentially** with distance
- **SLEM** is a fast early-design tool; fully coupled simulations should always be done for the final design
- **Common mode** (all bits in phase) and **differential mode** (target bit out of phase with neighbors) produce the **worst-case** impedance and velocity variations
- SLEM accuracy falls off when edge-to-edge spacing/height ratio is less than 1

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8. Crosstalk Trends

8.1 From Impedance Perspective

(chap3.pdf, p.36)

- **Lower-impedance traces** (wider trace or thinner dielectric) exhibit **less crosstalk-induced impedance variation**
- Lower-impedance lines confine the electromagnetic field more tightly to the ground plane → less coupling to adjacent signals
- Even when a line is not actively driven, its proximity to another trace **influences the impedance** of the active line (by increasing self-capacitance)

  Figure description (p.36): Graph of stripline impedance (ohms) vs. edge-to-edge spacing S (mils) for two line impedances (50 Ω and 65 Ω). Shows single-line Z_0, even-mode, and odd-mode curves diverging as spacing decreases. Higher-impedance lines show more variation.

8.2 From Mutual Inductance/Capacitance Perspective

(chap3.pdf, p.37)

- Mutual parasitics drop off **exponentially with spacing**
- Mutual inductance and capacitance for **lower-impedance lines are smaller** than for higher-impedance lines

  Figure description (p.37): Two graphs: left shows mutual inductance L_m (nH/inch) vs. edge-to-edge spacing for 50 Ω (lower) and 65 Ω (higher) lines — both decrease exponentially, with higher-impedance lines having larger L_m. Right shows same for mutual capacitance C_m (pF/inch) with similar trends.

8.3 Crosstalk Trends Summary

(chap3.pdf, p.38)

1. For a given dielectric constant, **low-impedance lines** produce **less impedance variation** from crosstalk
2. A single line's impedance is influenced by nearby traces even when they are **not actively driven**
3. Mutual parasitics fall off **exponentially** with trace-to-trace spacing

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9. Termination of Odd- and Even-Mode Transmission Line Pairs

9.1 Pi-Termination Network

(chap3.pdf, p.39)

Circuit: Two parallel lines with R_1 (line 1 to ground), R_2 (line 2 to ground), and R_3 (between the two lines) at the load.

Even mode ($V_1 = V_2 = V_e$, no current through R_3):

$$R_1 = R_2 = Z_{\text{even}}$$

(chap3.pdf, p.39)

Odd mode ($V_1 = -V_2 = V_o$, R_3 splits as $\frac{1}{2}R_3$ to virtual ground):

$$Z_{\text{odd}} = \left(\frac{1}{2}R_3\right) \| R_1$$

(chap3.pdf, p.39)

Solving for R_3:

$$R_3 = \frac{2(Z_{\text{even}})(Z_{\text{odd}})}{Z_{\text{even}} - Z_{\text{odd}}}$$

(chap3.pdf, p.39)

> **Figure description (p.39):** Pi-termination schematic. Two lines connect to R_1 and R_2 (each to ground), with R_3 bridging between the two lines. A differential receiver is connected across the two lines. Below: equivalent odd-mode circuit showing the virtual ground at the midpoint of R_3.

9.2 T-Termination Network

(chap3.pdf, p.40)

Circuit: R_1 in series on line 1, R_2 in series on line 2, R_3 between the midpoints of the two series resistors (connected to ground/reference).

Odd mode ($V_1 = -V_2 = V_o$):

$$R_1 = R_2 = Z_{\text{odd}}$$

(chap3.pdf, p.40)

Even mode ($V_1 = V_2 = V_e$, no current through R_3):

$$Z_{\text{even}} = R_1 + 2R_3$$

(chap3.pdf, p.40)

Solving for R_3:

$$R_3 = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}})$$

(chap3.pdf, p.40)

> **Figure description (p.40):** T-termination schematic. Series R_1 on line 1 and R_2 on line 2; R_3 bridges between the two series resistors to ground. Single-ended receivers at the output. Below: equivalent even-mode circuit showing $R_1 + 2R_3$ to ground.

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10. Minimization of Crosstalk

10.1 Rules of Thumb

(chap3.pdf, p.41–42)

1. **Widen the spacing** S between lines as much as routing allows
2. **Minimize** H (trace height above ground plane) while achieving target impedance — couples the trace tightly to ground, reducing coupling to adjacent signals
3. Use **differential routing** for critical nets (e.g., system clock)
4. Route signals on **different layers orthogonal** to each other to reduce inter-layer coupling
5. Route on **stripline or embedded microstrip** to eliminate velocity variations due to crosstalk
6. **Minimize parallel run lengths** between signals; avoid long coupled sections
7. Place components to **minimize trace congestion**
8. Use **slower edge rates** (with caution — has negative timing consequences)

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11. Additional Examples

11.1 8-Bit Bus Design Problem

(chap3.pdf, p.43–50)

Setup: - U1 driver: $Z_s = 30\,\Omega$, swing $= 0 \to 2\,\text{V}$ - PCB: 5 in. long traces, center-to-center spacing 15 mils, $Z_0 = 50\,\Omega$, $\varepsilon_r = 4.0$ - Stack-up: $W = 5\,\text{mils}$, $S = 10\,\text{mils}$, $T = 1.0\,\text{mil}$, $H = 3.2\,\text{mils}$ - Parasitics: $L_M = 0.54\,\text{nH/in.}$, $C_M = 0.079\,\text{pF/in.}$, $L_{11} = 7.13\,\text{nH/in.}$, $C_{11} = 2.85\,\text{pF/in.}$

(chap3.pdf, p.43–45)

Goal 1: Maximum impedance variation (common mode and differential mode)

Common-mode propagation (all 8 bits in phase):

$$Z_{2,\text{common}} = \sqrt{\frac{L_{22} + L_{12} + L_{23}}{C_{22} - C_{12} - C_{23}}} = \sqrt{\frac{7.13 + 0.54 + 0.54}{2.85 - 0.079 - 0.079}} = 55.26\,\Omega$$

(chap3.pdf, p.47)

$$\text{TD}_{2,\text{common}} = \sqrt{(7.13 + 0.54 + 0.54)(2.85 - 0.079 - 0.079)} = 148.6\,\text{ps/in.}$$

(chap3.pdf, p.47)

Differential-mode propagation (center bit out of phase with both neighbors):

$$Z_{2,\text{differential}} = \sqrt{\frac{L_{22} - L_{12} - L_{23}}{C_{22} + C_{12} + C_{23}}} = \sqrt{\frac{7.13 - 0.54 - 0.54}{2.85 + 0.079 + 0.079}} = 44.8\,\Omega$$

(chap3.pdf, p.49)

$$\text{TD}_{2,\text{differential}} = \sqrt{(7.13 - 0.54 - 0.54)(2.85 + 0.079 + 0.079)} = 135\,\text{ps/in.}$$

(chap3.pdf, p.49)

Results:

$$44.8\,\Omega < Z_o < 55.26\,\Omega$$

(chap3.pdf, p.49)

$$135\,\text{ps/in.} < \text{TD} < 148.6\,\text{ps/in.}$$

(chap3.pdf, p.49)

Goal 2: Maximum velocity difference

Velocity range corresponds to the TD range: 135 ps/in. to 148.6 ps/in.

Goal 3: Check for false triggering (common mode, overdriven)

Using the higher common-mode impedance ($Z_0 = 55.26\,\Omega$), with $Z_s = 30\,\Omega$:

$$V_{\text{initial}} = V_s \frac{Z_o}{R_s + Z_o} = (2)\left(\frac{55}{30+55}\right) = 1.29\,\text{V}$$

(chap3.pdf, p.50)

$$\rho_{\text{source}} = \frac{30-55}{30+55} = -0.29, \quad \rho_{\text{load}} = \frac{\infty - 55}{\infty + 55} = 1$$

(chap3.pdf, p.50)

Lattice diagram shows the load voltage starts at 1.29 V → jumps to 2.58 V → oscillates but stays above 1 V.

Conclusion: No false triggers since the voltage never drops below $V_{\text{TH}} = 1\,\text{V}$.

> **Figure description (p.50):** Lattice diagram showing V(source) and V(load) for common-mode propagation. The load initially jumps to 2.58 V, then rings above 1.0 V. A worst-case waveform at U2 shows the load voltage vs. time — the undershoot at $\sim 3 \times 10^{-9}$ s is circled as the concern, but it stays above 1 V.

Key Equations Summary

Quantity	Formula	Page
Inductive crosstalk noise	$V_{\text{noise}} = L_m \frac{dI}{dt}$	p.4
Capacitive crosstalk noise	$I_{\text{noise}} = C_m \frac{dV}{dt}$	p.5
NEXT current	$I_{\text{near}} = I(L_m) + I_{\text{near}}(C_m)$	p.11
FEXT current	$I_{\text{far}} = I_{\text{far}}(C_m) - I(L_m)$	p.11
Near-end magnitude (matched, long line)	$A = \frac{V_{\text{in}}}{4}\left(\frac{L_M}{L} + \frac{C_M}{C}\right)$	p.13
Far-end magnitude (matched)	$B = -\frac{V_{\text{in}} X \sqrt{LC}}{2T_r}\left(\frac{L_M}{L} - \frac{C_M}{C}\right)$	p.13
Mismatched victim adjustment	$V_x = V_{\text{crosstalk}}\left(1 + \frac{R - Z_0}{R + Z_0}\right)$	p.15

Quantity	Formula	Page
Coupling factor K	$K = L_{12}/\sqrt{L_{11}L_{22}}$	p.21
Odd-mode inductance	$L_{\text{odd}} = L_{11} - L_{12}$	p.25
Odd-mode capacitance	$C_{\text{odd}} = C_{11} + C_m$	p.26
Odd-mode impedance	$Z_{\text{odd}} = \sqrt{(L_{11} - L_{12})/(C_{11} + C_m)}$	p.27
Even-mode inductance	$L_{\text{even}} = L_{11} + L_m$	p.28
Even-mode capacitance	$C_{\text{even}} = C_{11} - C_m$	p.29
Even-mode impedance	$Z_{\text{even}} = \sqrt{(L_{11} + L_{12})/(C_{11} - C_m)}$	p.29
Pi-termination center resistor	$R_3 = 2Z_{\text{even}}Z_{\text{odd}}/(Z_{\text{even}} - Z_{\text{odd}})$	p.39
T-termination bridge resistor	$R_3 = \frac{1}{2}(Z_{\text{even}} - Z_{\text{odd}})$	p.40